Catherine Lin's paper tackles a timely and important topic: how artificial intelligence is reshaping the skills, roles, and job opportunities for computer scientists across different career stages. The research question is clearly framed, and the structure of the paper is logical, moving from a background discussion of AI, AGI, and LLMs, to a detailed analysis of workforce implications across entry-level, mid-level, PhD/researchers, and senior professionals
. The inclusion of both literature review and data from a Kaggle dataset lends credibility, and the use of evaluation dimensions (required skills, job market access, AI's impact, and need for up-skilling) provides a consistent framework for comparison.

One of the strengths of the paper lies in its comprehensive coverage. The author surveys a wide range of industry reports, academic papers, and workforce data, synthesizing them into a cohesive analysis of shifting job market dynamics. Particularly strong is the integration of short-, mid-, and long-term projections, which shows awareness of how fast-moving technologies create both immediate pressures and long-range uncertainties
. The comparative section and recommendations are also well considered, giving practical takeaways for different experience levels.

That said, the paper would benefit from sharper focus in places. At times, the narrative repeats similar points across groups (e.g., the importance of AI literacy, governance, and evaluation), which could be streamlined to avoid redundancy. Some sections could use tighter editing for clarity and grammar, as awkward phrasing occasionally distracts from the argument. The dataset analysis, while creative, is illustrative rather than definitive, and the paper should emphasize its limitations more explicitly to avoid overstating claims. More critical evaluation of potential counterarguments—for example, the resilience of certain human skills that AI cannot easily replicate—would strengthen the discussion.

Overall, this is an ambitious and well-researched paper for a high school scholar. It demonstrates intellectual curiosity, the ability to synthesize diverse sources, and a serious effort to structure an argument in the style of academic research. With some refinement in writing, deeper critical engagement with sources, and a clearer distinction between speculation and evidence, the paper would be even stronger. It is a commendable effort that reflects both strong preparation and promising research potential.

How is AI transforming job opportunities, roles, and skill demands for computer scientists of varying experience levels, from the Pre-AI period to the present and into the future?

Catherine Lin

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August 28, 2025

Abstract

Today, the public is on high alert through the media that "AI will replace all computer scientist", however, growing evidence shows that AI is reshaping rather than erasing the field. This paper analyzes four different workforce readiness dimensions, and how AI transforms them for computer scientists at four experience levels: entry-level, mid-level, PhD/researchers, and long-time professions across three time periods from Pre-AI to present and into the future split into 3 sub-periods: short-term, mid-term, and long-term. The four dimensions of workforce readiness include: required skills, job market access, AI's impact, and the necessity for up-skilling are evaluated through the combination of literature reviews, industry reports, and dataset with directional properties. A comparison is also made throughout the workforces summarizing how the different dimensions evolve over time.

Research indicates that AI values are shifting from manual structured coding towards practices related with specification, management, evaluation, and governance. Entry-level access to the workforces tightens as low-complexity tasks diminishes, increasing the priority on verified AI use, testing, and documentation. Mid-level roles are benefit when candidates demonstrate a level of AI-awareness in delivery and reliability techniques. PhD and researchers roles surrounds around evaluation on science, data oversight, and applied work. Long-time professions concentrates on a product ownership, LLMOps oversight, and different risk management upon AI use. Overall, AI redefines the careers workforce readiness criteria and advances different role at various scales, but one central idea is clear: human oversight remains key to developing credibility in the workforce.

Keywords: Artificial Intelligence (AI), Artificial General Intelligence (AGI), Large Language Models (LLMs), workforce readiness, software labor markets, orchestration, LLMOps, evaluation, governance

Contents

1 Introduction **4**
- 1.1 Rise of AI in the Workforce . . . 4
- 1.2 Research Question and Purpose . . . 4

2 Related Work **5**
- 2.1 2024 Stack Overflow Developer Survey . . . 5
- 2.2 The Impact of AI on Developer Productivity: Evidence from GitHub Copilot . . . 5
- 2.3 Generative AI at Work (NBER Working Paper 31161) . . . 5
- 2.4 Developer Productivity with Generative AI . . . 5
- 2.5 World Economic Forum: Future of Jobs Report 2023 . . . 5
- 2.6 Engineering for Machine Learning: A Case Study . . . 5

3 Background **5**
- 3.1 What is Artificial Intelligence (AI) . . . 5
- 3.2 What is Artificial General Intelligence (AGI) . . . 6
- 3.3 What are Large Language Models (LLMs) . . . 6
- 3.4 Comparing Them All Together . . . 6
- 3.5 Why Focus on Computer Scientist Specifically? . . . 6

4 AI's Role in the Software Job Market **6**
- 4.1 Productivity Booster . . . 6
- 4.2 Depletion of Easy Work with New Areas of Focus . . . 7
- 4.3 AI Basics Become a Required . . . 7

5 Impacts of Different Groups in the Workforce **7**
- 5.1 Section 5: Introduction and Format . . . 7
- 5.2 Dataset Description and Scope . . . 8
- 5.3 Entry-Level / Beginner Coders . . . 8
  - 5.3.1 Entry-Level Required Skills . . . 9
  - 5.3.2 Entry-Level Necessity for Up-skilling . . . 9
  - 5.3.3 Entry-Level Job Market Access . . . 10
  - 5.3.4 Entry-Level AI's Impact . . . 11
- 5.4 Mid-Level Coders (College Graduates) . . . 12
  - 5.4.1 Mid-Level Required Skills . . . 13
  - 5.4.2 Mid-Level Necessity for Up-skilling . . . 14
  - 5.4.3 Mid-Level Job Market Access . . . 15
  - 5.4.4 Mid-Level AI's Impact . . . 16
- 5.5 PhD Students / AI Researchers . . . 16
  - 5.5.1 PhD Students / AI Researchers Required Skills . . . 18
  - 5.5.2 PhD Students / AI Researchers Necessity for Up-skilling . . . 19
  - 5.5.3 PhD Students / AI Researchers Job Market Access . . . 20
  - 5.5.4 PhD Students / AI Researchers AI's Impact . . . 20
- 5.6 Long-Time Professionals / Managers . . . 21
  - 5.6.1 Long-Time Professionals / Managers Required Skills . . . 22
  - 5.6.2 Long-Time Professionals / Managers Necessity for Up-skilling . . . 23
  - 5.6.3 Long-Time Professionals / Managers Job Market Access . . . 24
  - 5.6.4 Long-Time Professionals / Managers AI's Impact . . . 24

6 Compare and Contrasting the Different Field Together **26**
- 6.1 Values AI is Changing . . . 26
- 6.2 New Roles Created . . . 27
- 6.3 Labor Polarization . . . 27

7 Discussions and Reflection **28**
7.1 Views . . . 28
7.2 Influence on Career Pathway . . . 28
7.3 Recommendations for The Workforce . . . 28
7.3.1 Suggestions for Entry-Level . . . 28
7.3.2 Suggestions for Mid-Level . . . 28
7.3.3 Suggestions for PhD and Researchers . . . 28
7.3.4 Suggestions for Long-Time Professionals . . . 29

8 Future Works **29**
8.1 Building Better Datasets . . . 29
8.2 Future Analysis . . . 29

9 Conclusion **29**
9.1 Summary of Affected Groups . . . 29
9.2 Final Insights and Message . . . 30

1 Introduction

In today's society, you can open any feed only to see the same headline: "AI will replace computer scientists". These news and articles raise great panic in the workforce with students majoring in computer science losing sight in their future to professions that have been in the workforce for years praying on surviving the new rounds of layoffs. These messages work like the 21 century snake oil. Snake oil is a 19th century medication that is promoted to cure all illnesses.[1] However, AI doesn't seem as powerful as it may seem.

1.1 Rise of AI in the Workforce

Artificial intelligence, especially large language models like ChatGPT, Copilot, and Claude, has developed rapidly from experimental tools to now, everyday foundation within the workforce. In just a few years, generating code, testing , documenting, and handling data have become AI automated by default in many teams and companies. However, the public has split into two party: one which people predicts AI's sweeping job replacement, while another sends warnings of the "AI snake oil," or the exaggerated claims that over the internet. This warns people to judge by evaluations and real data rather than trusting hypes and scenarios.

At the same time, anxiety about jobs is viewed across the workforce. Reviews show a mass amount of workers expect AI to reduce opportunities over the long run as many acknowledge AI's usefulness and impact in speeding up basic tasks.[2] Within some companies, adoption is already seen reshaping expectations. Employers start to look for "AI-aware" candidates who are able to use assistants, verify their outputs, and integrate tools into workflows. Recent workforce data indicate that most knowledge workers are now using AI at work, and many leaders report that they would not hire candidates who lack AI skills. This showcases thate AI is becoming a baseline or standard, and not a something extra. Other surveys show that a strong understanding of AI coding tools, people will not do not direct view it as a threat to their own jobs, since concerns about accuracy, security, and ethics are brought up. All in all, these components form a change in what is valued: less time on manual standard work, and more attention on reviewing and big-picture thinking.

1.2 Research Question and Purpose

This paper asks: How is AI transforming job opportunities, roles, and skill demands for computer scientists of varying experience levels, from the Pre-AI period to the present and into the future? The analysis focuses on the workforces consequences like hiring, roles, and required skills. By examining surveys, industry reports, and literature papers, the paper evaluates how AI affects four workforce groups differently. The goal is to separate reliable proven trends from the public hype, and clarify where demands are rising or falling, and identify skills that remain resilient even as AI automation advances.

The structure of the article is broken up into different pieces. The article start of strong with a summary of the works related in Section 2. Section 3 provides the background on AI, AGI and LLMs and explains why the subject group is examined. Section 4 gives an review over the evidence on AI's effects in the job market, including the different productivity and hiring expectations in various sections. Section 5 analyzes impacts by the 4 different work work groups while both highlighting the risks and opportunities. Section 6 makes an comparison across different major ideas such as shifting skill value, labor polarization, and governance. Section 7 provides an authors view of the workforce and strategies for each workforces success. Section 8 provides prompts of some future works. Lastly, section 9 end with a wrap-up of the whole analysis.

2 Related Work

2.1 2024 Stack Overflow Developer Survey

Large, recurring survey of developers that tracks tool adoption, hiring signals, and perceived skill gaps. The 2024 edition reports rapid uptake of AI assistants alongside persistent concerns about accuracy and evaluation, with differences by experience level. Findings directly inform trends in required skills and job-market access across groups.[3]

2.2 The Impact of AI on Developer Productivity: Evidence from GitHub Copilot

Empirical study showing measurable speed gains on coding tasks with AI assistance, with task type and developer experience moderating effects. Results support the view that teams can deliver similar output with fewer manual hours, shifting skill demand toward testing, review, and integration.[4]

2.3 Generative AI at Work (NBER Working Paper 31161)

Randomized deployment of genAI in customer support shows the largest productivity lift for less-experienced workers, compressing performance gaps. The mechanism, formalized know-how exposed via AI has clear implications for software roles: entry-level ramp-up accelerates while differentiation moves to system design, assurance, and governance.[5]

2.4 Developer Productivity with Generative AI

Industry analysis and case studies quantifying time reallocation and outlining practices needed to convert individual tool use into team-level gains. Argues that role definitions will emphasize orchestration and evaluation, informing mid-level and senior expectations.[6]

2.5 World Economic Forum: Future of Jobs Report 2023

Global employer survey projecting job creation or displacement and the skills most in demand through the near term. Identifies AI literacy, analytical thinking, and systems integration as rising priorities, providing macro context for cohort-specific findings on required skills.

2.6 Engineering for Machine Learning: A Case Study

Seminal case study on how ML systems change engineering practice. Explains why required skills for developers and researchers broaden beyond coding into experimentation infrastructure and life-cycle assurance, central to the paper's skill-shift argument.[7]

3 Background

Before we even dive into the logistics of artificial intelligence (AI). We first have to fully understand what artificial intelligence (AI) is, what it is different from artificial general intelligence (AGI), and the large language model (LLM).

3.1 What is Artificial Intelligence (AI)

Abbass states that Artificial Intelligence is a machine, integrated into the society, that performs competitive tasks using cognitive processes and communication with the society through social and cognitive phenomena of the society. [8] In simpler terms, it is a field that builds machines to perform or replicate tasks that require human intelligence, such as reasoning, learning, or logical thinking. Today, AI has been developed to perform well-rounded automation tasks like data processing, natural language and vision, and making optimal decisions while in the same

time continuously improve and adapt, learning how to strengthen itself. However, with capabilities comes limitation, AI lacks the general intelligence and consciousness that humans do. Additionally, AI systems contain great amount of bias as it depends on a large amounts of data sets which may lead to unfair decisions. Currently, many individuals are still uncertain about AI use having concerns over its accountability and safety hazards.

3.2 What is Artificial General Intelligence (AGI)

According to Xu, "An Artificial General Intelligence (AGI) is a computer system that is adaptive to its open environment with limited computational resources and that satisfies certain principles of universality and autonomy."[9] In simple terms, AGI is a system that can handle any unexpected or real-world simulations by adjusting its behavior performed with flexibility and independency. The difference between AI and AGI is that AGI has the ability to adapt to new situations, and able to approach problems that it has never encountered; this allows it to perform tasks more like humans. Unfortunately, current AI system don't show any standards to meet the values for AGI, meaning that it doesn't exists in the public, yet. [10]

3.3 What are Large Language Models (LLMs)

Large language models (LLMs) are machine learning systems trained through the input of vast amounts of text and datasets to predict and generate human-like language. [11] In simpler terms, LLM's are machine systems that learn from a vast amount of text to they can predict or guess the next output to give based on the input in hopes to sound like it was written by a human. LLM's can generate related information to the topic, summarize information, and perform tasks that related to retrieving data, however, LLM's lack the ability to process like humans, have high error rates, and may output misinformation that are non-existent.[12]

3.4 Comparing Them All Together

To grasp a better understanding of the relationship between the 3, image a kitchen. Within a kitchen, AI would be a single appliance, like a toaster or blender, which is great for one specific task or job. LLMs is a fast recipe that is auto-completed, predicting and suggesting steps and ideas for the dish, without guarantee of quality or credibility. AGI is like the master chef who can plan menus, cook any kind of cuisine, and manage the kitchen with ease.

3.5 Why Focus on Computer Scientist Specifically?

This article focuses on computer scientists as they are the first occupations to be impacted by AI changing day-to-day work and skills. Code is like text, and many recent models trained in code shows ability to develop code that requires years of training.[13] Hiring changed here first, too, as interviews and code reviews now expect candidates with AI awareness. Moreover, adoptions of AI in this field are increasing rapidly as studies show that developers using AI tools have rose from 62 to 84 percent.[4] The great news and AI's rapid replacements of jobs in this field has planted fear in many students with interest to pursue this field. Studying this field provides evidence over the hype that AI will replace computer scientist, and give an overview of what may be coming for other technical roles.

4 AI's Role in the Software Job Market

4.1 Productivity Booster

Different AI assistants has cut down the time needed for many routines software task like: developing standardized setup code, units tests to check functions, and short write-ups of the purpose and basic usage of code. Developers using AI code tools have shown a great decrease in total amount of time needed to finish task and projects in both experiments and real teams.Studies

show that lab-style tests had an estimate of 55 percent increase in the time of completion. Similarly, real-life teams in companies have seen a 50 percent sooner to pull request merge which are proposals to mere code changes from a fork into the main branch of code.[14][15] Beyond speeding up coding process, evidence also show that AI helps raise productivity the most for of the less experienced workers, by adding pressure and hints of consistent of skilling and leveling up. AI acts like a warning to workers to keep improving themselves to ensure they don't get replaced by AI which increases the stress, productivity, and development in the field.[5][16]

4.2 Depletion of Easy Work with New Areas of Focus

As low-complexity tasks are completed by AI, team roles shift toward composition like: connecting components, writing and implementing tests, reviewing AI-generated code, and ensure security measures are in place. [17] New AI powered tools are capable to take action on their own, like submitting pull requests for fixing small errors.[18] As a result, leaders from large corporations state that AI will cause the restructure of hiring plans, enforcing smaller teams for easy task and higher expectations for code monitor and reviews.[19][20]

4.3 AI Basics Become a Required

Company employers are looking for applicants who are AI-aware able to: verify and debug AI outputs, connect with AI assistants safely with backend, prompt clearly, and following privacy rules. Markets now show a trend of hiring applicants with AI related skills. With AI taking over mass amount of work once done by human, one rule is clear: review everything AI produces. [21]

5 Impacts of Different Groups in the Workforce

Within this section, an evaluation of different groups in the workforce will be analyzed in four different workforce readiness dimensions through three different time periods. Adding on, an analyze of a dataset from Kaggle will also be included to provide statical analyze and further support for two of the dimensions with in the workforce readiness. [22] Below I will Introduce how this sections is formatted, how the dataset was organized, and, lastly, the analyze for each workforce group.

5.1 Section 5: Introduction and Format

In this section, I will be analyzing computer scientists into 4 main categories: entry-Level, mid-Level, PhD students or researchers, and long-time professionals (see Figure 1). For each workforce category, I will evaluate the changes in workforce readiness, specifically examining the four dimensions: required skills are the skills employers expect job applicants to already know and perform in the job, the necessity for up-skilling is how often workers have to learn new tools or methods to stay effective and employable, job market access is how easy it is to get an interview or offer, and AI's impact is how much AI changes or changed one's task, productivity, team size, and evaluation for a specific group. As you can see in figure 1 each workforce readiness is then further analyzed in 3 time periods: Pre-AI, Present, and Future with 3 terms. The different time periods are determined on the basis of how fast AI tools appearedand the turning points in the past and estimated in the future. For the purpose of this article, Pre-AI is identified as any year that comes before 2022, present is the years between 2022 and 2025, and future with short-terms from 2025 to 2030, mid-terms from 2030 to 2040, and long-terms from 2040 and beyond, also visible with the timeline in figure 1. Each category will also include a table with a rating scale of 1-5, where 1 represents low difficulty for required skills and access to the job market, urgency for the need for up-skilling, and minimal impact for the impact of AI, and 5 represents high difficulty for the required skills and job market access, urgency for the necessity of up-skilling, and maximum impact for the impact of AI, to ensure a consistent comparison between the different components and periods of time.

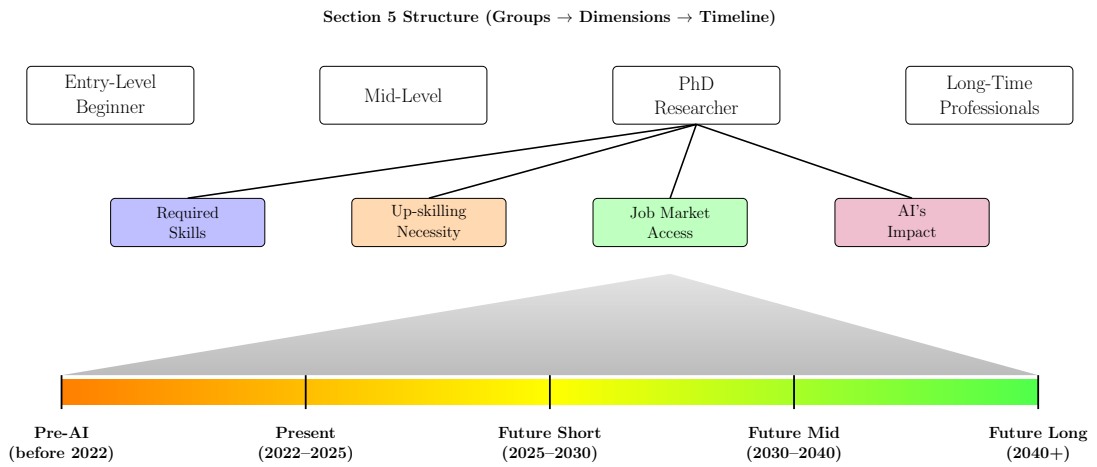


Figure 1: This is how Section 5 is sectioned. This figure illustrates how each group is analysed in each workforce dimension in the different time periods.

5.2 Dataset Description and Scope

This study utilizes the AI Impact on Job Market: Increasing vs Decreasing Jobs (2024–2030) dataset sourced from Kaggle.[22] Please note that this is an estimated dataset compiled from public job data patterns sourced from: U.S. BLS, OECD, McKinsey, and WEF reports; all sources are either primary statistics, peer-reviewed papers, and high-quality industry reports. However, the final dataset is illustrative, and not official statistics. Lastly, this data set will be used mainly to analyze present and future components, while literature reviews are used to analyze Pre-AI components.

This data set includes 30000 different job records across multiple industries including attributes such as job titles, required educations, AI impact, years of experience, automation risk, and estimation of projected job openings from 2024 to 2030.

To filter and prepare the dataset of this study I used Python with Pandas library to automatically organize the data. First, I filtered out all the IT category jobs into a file. Next, within the a new file each I filtered out all the jobs that need a high school, associates, and bachelors degree with 0 to 3 years of experience; masters degree with 0-7 year of experience; and Doctors degree with 0-10 years of experience into separate files. Lastly, I filtered out all the jobs with the IT field that needed 10 or more years of experience into a file.

Four addition variables will be analyzed to reflect one or more of the components listed in the beginning of this section. AI impact level which estimates level of AI impact on the job in 3 dimensions: low, medium, high to look into AI's impact. Job status which indicates if the job is increasing or decreasing due to AI adoption to look into job market access. Job openings presenting the number of jobs open in 2024, and projected openings by the year 2030 to both also give insights to the job market access.

The data will be used when analyzing for each workforce groups job market access and AI's impact during the present and into the future.

5.3 Entry-Level / Beginner Coders

Before we start, a clear definition or boundary needs to be set for the group we are talking about. For the purpose of this research article entry-level or beginner coders are mostly undergraduates that are entering the job market for the first time. Historically, this group has faced the greatest setbacks in entering the workforce due to the constrained amount of skills, limited experience, inflexible working hours, and an increasing competition due to the majories increasing popularity. Since these group of people are still learning concepts in college, their skills are very limited

including: simple knowledge of algorithms, debugging, proficiency in some or all popular coding languages, and many more surface level qualities. With limited skills in-hand, the mass majority of undergrads would aim to find an entry-level job.

5.3.1 Entry-Level Required Skills

During Pre-AI, when AI wasn't very accessible to the public, only basic technical skill were needed like: programming (java, python,etc), data management, project management, and so on. Although these technical skills are necessary for the productivity and success of a company, many valued soft skills, which are basic skills such as communication, teamwork, problem solving, etc., had greater value than technical ones.[23] Overall, as shown in Table 1, the difficulty of the skills required during the Pre-AI period is rated 2.

Unlike Pre-AI, present-day entry-level expectations of skills have started to shift rapidly do to the rising technological advancements, AI. Since many of the technical skills, like GitHub Copilot automating basic coding tasks, can be replaced by AI, now applicants are demanded to be proficient in AI related tools and shift their focus to understand higher technical skill than the past. [**?**] Soft skills are also a must in any position or workspace. As shown in Table 1, the difficulty of the skills required in the present is rated 3. In the short term, technical skills have evolved drastically in recent decades; however, the core fundamentals have never changed.[23] But will they change in the future is another question to focus on. One thing is for sure, in the future, you need to have a very good program understanding very early on. [24]

In the short-term future, from 2025 to 2030, many skills will be aligned with AI. Developing the abilities to co-code and communicate efficiently with LLM's, like ChatGPT, and having a strong understanding of AI ethics and bias are all important skills to acquire. That means that employers will look for applicants that have defensive habits that can sense insecure code that are generated by these AI assistants.[25] For example, the road mapping for software engineering emphasizes human-in-the-loop verification, meaning that a person must review, correct, and approve AI's work.[26] With evaluation, short-term future is ranked 4 as the difficulty of skills required, shown from table 1.In the mid-term future, from 2030 to 2040, we will evolve even more advanced AI skills. Skills includes: interdisciplinary knowledge enabling the connection of coding to other advanced technology fields with AI systems, guide and supervise AI development, and customize AI models. That is, the roles will develop into an evaluation and understanding of an orchestrate of agentic workflows and tools.[26] Since hallucinations will remains which requires stronger evaluation skills to validate code produced by AI, the baseline skills will include Large Language Model Operations (LLMOps), which refers to the specialized practices and tools used to manage the entire generative cycle of large language models (LLMs) in production.[27][28] Overall giving a ranking of 4, as shown from table 1. Lastly, looking into long-term, beyond the year of 2040, entry-level works would likely shift toward specification where they would format clear and precise specs that AI turns into verified code.[26] This means that entry level groups should have acquired skills or ways to prove that the AI generated code actually does what it is suppose to do. As shown from table 1 the difficulty of the skills required during the future long-term period is rated a 5, the highest of them all.

In general, though the Pre-AI stage may have been easier for undergrads to land a entry-level job than the present and future, universities failing to align their curriculum to the evolving industry needs, especially after the introduction of AI, adds an extra level of difficulty for students seeking a job.[29] Nevertheless, as AI's abilities increases and the foundations of AGI grows more mature the skills required or this entry level workers will sky rocket in difficulty parallel as well. The ranking of difficulty of required skills shown in table 1 increases as the time increases creating a direct relationship.

5.3.2 Entry-Level Necessity for Up-skilling

Within the Pre-AI era up to the end of 2021, beginners mainly built core software engineering skills such as requirements writing, software testing, and version control, which means keeping track of changes to code over time so that nothing is lost.[30] Teams also practiced DevOps, which means combining development work with operations work to ship software faster and

Table 1: Entry-Level Coders: Workforce Impact Evaluation Scale

Dimension	Pre-AI	Present	Future Short-term	Future Mid-term	Future Long-term
Required Skills	2	3	4	4	5
Job Market Access	4	3	3	2	2
AI's Impact	1	3	4	5	5
Necessity for Up-skilling	2	4	4	5	5

Note: Pre-AI = before 2022; Present = 2022–2025; Future Short-term = 2025–2030; Future Mid-term = 2030–2040; Future Long-term = 2040 and beyond.

safer, and SRE or site reliability engineering, which is a discipline that keeps services reliable through automation and careful monitoring.[31] Artificial intelligence tools were not widely used in day to day student workflows, so AI specific study was limited. For this period the urgency to up-skill in AI related areas is low and the rating is a 2, as you can see in Table 1.

In the present between the years 2022 and 2025, AI coding assistants are common, so beginners need AI literacy, which means understanding what an AI can and cannot do, how to write a clear prompt, and how to check the output for mistakes.[3] A prompt is a precise instruction given to an AI system, and good prompting plus code review helps catch errors early.[4] Many teams also expect basic safety habits such as simple security checks and fairness checks when data is involved. For this period the urgency to up-skill is high and the rating is a 4, as you can see in Table 1.

Within the short term future between the years 2025 and 2030, most developers are expected to use AI tools in daily work, so beginners need to learn them the way earlier generations learned version control. Retrieval augmented generation, often called RAG, will matter because it means the AI looks up facts from trusted sources before answering so that hallucinations are reduced.[**?**] Simple compliance habits will spread as well, which means keeping records about what data was used and how results were produced. For this period the urgency to up-skill is extremely high and the rating is a 5, as you can see in Table 1.

Within the mid term future between the years 2030 and 2040, work shifts from only typing code to running AI features safely in production, which many teams call LLMOps and that means the day to day operation of large language model features including monitoring, testing, and safe rollback when something goes wrong.[32] Beginners will need to understand observability, which means the ability to see system health through logs, metrics, and traces, and auditability, which means the ability to trace and explain how a result was produced and who changed what so that an independent person can check it.[33] Organizations will grow formal AI management systems, which means policies and processes that manage AI risks. For this period the urgency to up-skill stays at the maximum and the rating is a 5, as you can see in Table 1.

Within the long term future beyond the year 2040, tools will turn clear specifications into working code more often, so beginners will stand out by writing precise specifications, which are exact descriptions of what software must do, and by building tests that prove the behavior is correct.[6] Even in entry roles, regulated sectors will expect careful documentation and full audit trails so that decisions can be reviewed later. Continuous learning will be necessary because the tools and the rules will keep changing. For this period the urgency to up-skill remains at the maximum and the rating is a 5, as you can see in Table 1.

Across the five periods, the center of gravity moves from basic software craft in the Pre-AI era to constant AI supported work after the year 2025. The skills that protect employability are prompt clarity, verification, simple governance habits, and the ability to run AI features safely, not only the ability to write code. The overall message is that the earlier a student builds AI literacy and testing discipline, the easier it becomes to meet the rising expectations in the short term, the mid term, and the long term.all

5.3.3 Entry-Level Job Market Access

In Pre-AI (Before 2022), entry screening emphasized data structure interviews, course projects, and internships rather than AI tool fluency, producing a competitive yet tractable funnel into

junior roles.[34] In that period, employer outlooks discussed automation broadly but did not elevate AI literacy to a baseline prerequisite for interview access at the entry stage. As a result, signaling through internships, open-source contributions, and capstone projects dominated interview selection effects rather than proficiency with copilots or prompting workflows. This corresponds to an access-difficulty rating of 3; see Table 1.

In the Present (2022–2025), developer and workplace surveys show rapid diffusion of AI in daily practice about 62 percent of respondents report currently using AI tools and 76 percent use or plan to use them, which raises interview expectations for AI-augmented workflows even at the first rung.[3] Referencing the Kaggle graphs, the Entry status bars indicate more titles labeled "decreasing" than "increasing"; Fig. 2), and the openings curve points downward toward 2030 (Fig. 4), both of which imply a tighter near-term funnel. The same adoption pattern at the workplace level 75 percent of global knowledge workers using AI suggests that interviews increasingly probe orchestration, verification, and AI-aware problem decomposition rather than manual output alone. This corresponds to an access-difficulty rating of 4; see Table 1.

In the Short-Term Future (2025–2030), the Kaggle slope graph shows approximately a 5.5 percent decline in Entry openings by 2030; Fig. 4), consistent with assistants absorbing low-complexity tasks and reducing net junior requisitions. Macro assessments project that generative AI will keep lifting productivity on the order of 0.1–0.6 percentage points in labor-productivity growth annually through 2040 if adoption and redeployment proceed which typically compresses rote work while rewarding candidates who can supervise and integrate AI safely.[6] In parallel, workplace tracking indicates sustained mass adoption, reinforcing that baseline AI literacy will be expected in screening. This corresponds to an access-difficulty rating of 4 through 2030; see Table 1.

In the Mid-Term (2030–2040), employer outlooks anticipate reallocation of tasks and significant re-skilling rather than uniform job loss, opening paths for AI-literate entrants into QA, integration, data hygiene, and compliance-aware support where human oversight remains crucial.[34] Broader macro analyses argue that productivity gains can stabilize entry opportunities when organizations invest in training and governance that redeploy time savings to higher-value activities.[6] As these practices standardize, interview criteria are expected to prioritize evidence of AI-first workflows and safe tool stewardship alongside fundamentals.[34] This corresponds to an access-difficulty rating of 3 for 2030–2040; see Table 1.

In the Long-Term (2040+), agentic systems are expected to be embedded across the delivery stack, and entry pipelines will emphasize orchestration, verification, and safety over boilerplate production, shifting interview emphasis toward demonstrable judgment with AI tools.[34] As AI-augmented curricula and credentials mature, thresholds are likely to remain elevated but clearer and more standardized for applicants who can document these competencies. This corresponds to an access-difficulty rating of 3 in the long run; see Table 1.

In Conclusion, the combined signal from the Kaggle figures, a slight tilt toward "decreasing" titles and a downward openings trend by 2030 (Figs. 2, 4), aligns with external evidence of widespread AI adoption and near-term crowding, followed by stabilization as AI-ready pathways formalize. Overall, the trajectory for Entry job-market access is down then flat, corresponding to a present rating near 4 that moderates to 3 beyond 2030; see Table 1.

5.3.4 Entry-Level AI's Impact

In Pre-AI (before 2022), entry-level work centered on boilerplate, bug-fixes, and toolchain familiarization, with minimal direct influence from AI on productivity expectations or evaluation criteria in most teams.[34] Mentorship and code review norms rather than AI prompting typically governed the ramp-up for novices during this period. This corresponds to an AI-impact rating of 1 and an upskilling-urgency rating of 1; see Table 1.

In the Present (2022-2025), controlled experiments report 55.8 percent faster task completion with GitHub Copilot on standardized programming tasks, materially shifting expectations for throughput and initial code scaffolding.[4] Workplace evidence concurrently shows 75 percent of knowledge workers using AI, which embeds prompting, tool chaining, and AI-assisted test design into ordinary development practice. The Kaggle impact bars for Entry lean Moderate/High (Fig. 3), consistent with evaluation moving toward AI-aware debugging, documentation, and

verification. This corresponds to an AI-impact rating of 4 and an upskilling-urgency rating of 4; see Table 1.

In the Short-Term Future (2025–2030), assistants become default tools across the toolchain, so entry roles concentrate on supervising model outputs, strengthening tests, curating examples/data, and maintaining CI/data quality even as openings dip (Fig. 4). Macro projections indicate ongoing productivity gains from generative AI if organizations redeploy saved time effectively, amplifying the premium on orchestration over manual keystrokes.[6] Given these conditions, evaluation is expected to weigh prompt design, failure-mode anticipation, and safety checks as core competencies.[34] This corresponds to an AI-impact rating of 4 and an upskilling-urgency rating of 5 through 2030; see Table 1.

In the Mid-Term (2030–2040), rising autonomy shifts more scaffolding and maintenance to AI, reorienting human contribution toward supervision, traceability, compliance, and socio-technical fit that cannot be fully automated.[34] With these shifts, entry contributors are evaluated less on raw output volume and more on system-level reliability and safe deployment practices. This corresponds to an AI impact rating of 5 and an urgent capacity upgrade rating of 5 for 2030–2040; see Table 1.

In the long term (2040+), the saturation of AI across the delivery leaves the human role at the beginning most valuable in judgment-intensive tasks, security, privacy, and governance-aware changes, while routine generation and refactoring are increasingly automated.[34] Continuous learning remains central as benchmarks and agent capabilities evolve, keeping early career expectations dynamic. This corresponds to an AI-impact rating of 5 and an upskilling-urgency rating of 5; see Table 1.

In Conclusion (Impact), experimental and workplace evidence indicates that AI's influence on entry-level software work rises steeply, with performance increasingly judged on orchestration, verification, and responsible use rather than manual code volume, which aligns with the Kaggle impact distribution (Fig. 3). Overall, the impact trajectory is a strong incline, corresponding to sustained 5 impact and upskilling-urgency ratings across future horizons; see Table 1.

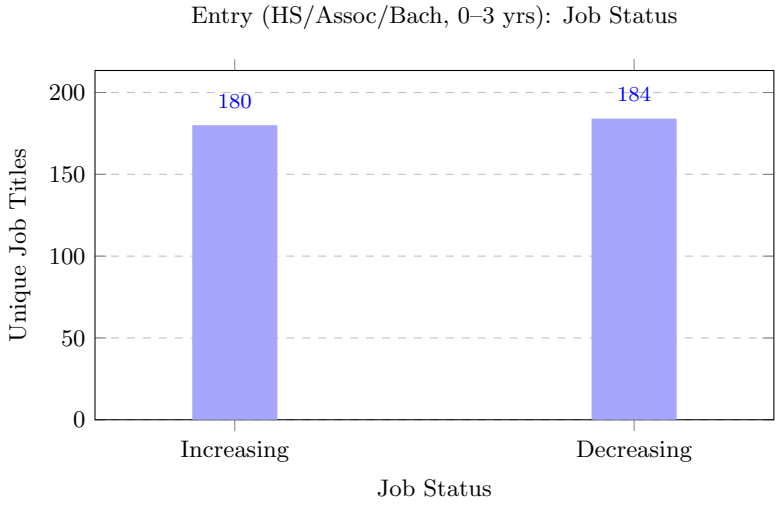


Figure 2: Entry — unique job titles labeled Increasing vs Decreasing.

5.4 Mid-Level Coders (College Graduates)

In this paper, mid-level coders are identified as early-career professionals with roughly around 3-7 years of experience, most of the time graduates, added on with advanced interns or juniors who already step foot into and contributed to the workforce. Most of the time their role is to build features that are end-to-end on medium level tasks, fixing bugs, write tests, and help keep services running with the guidance from seniors in the workforce. A common stereotype is that they're "just implementers" who follow commands and tasks without big picture thinking.

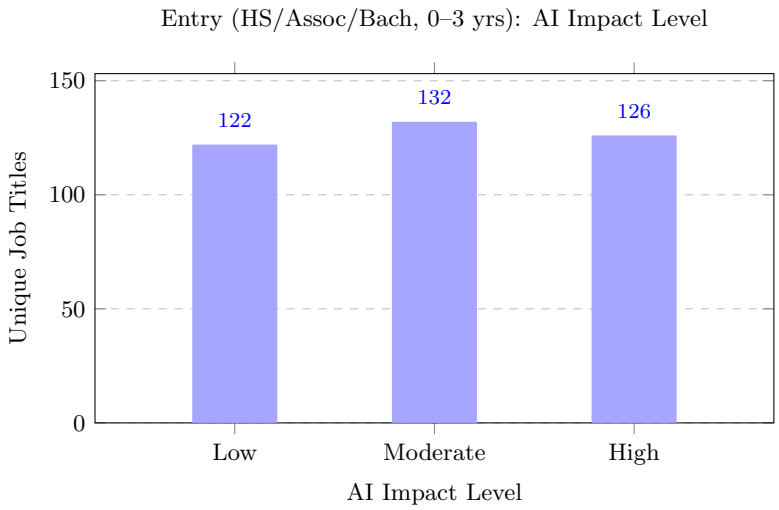


Figure 3: Entry — unique job titles by AI impact level.

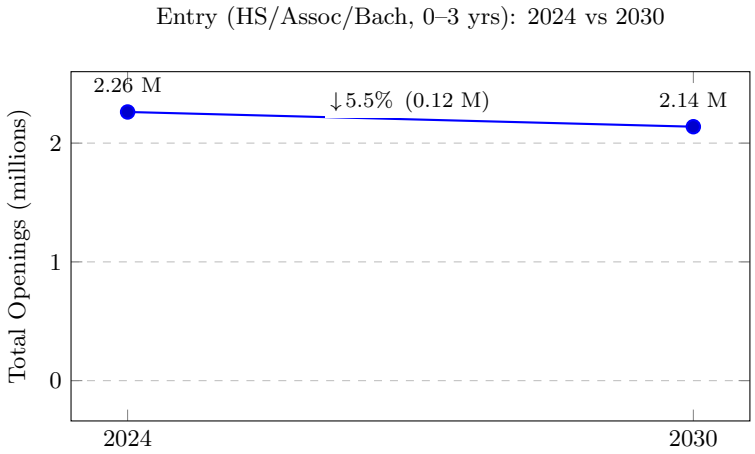


Figure 4: Entry — projected openings decrease slightly from 2024 to 2030.

Competition is often intense from global hiring and AI tools that speeds up coding routines, and limits includes fewer mentorship slots, pressure to learn new tools constantly, and a need to prove that they can define, drive, and deliver solutions, not just produce code.

5.4.1 Mid-Level Required Skills

Before widely available LLM tools and open access to AI, mid-level developers were often hired for solid software engineering fundamentals. Employers were expected to be fluent and capable to perform skills across the classic SWEBOK, an abbreviation of requirements, design, construction, testing, maintenance, and configuration management, knowledge areas and the ability to work in teams using engineering and software processing like version control, code review, integration testing, and issue tracking. [35] Research dated on and before the 2021, issued that many “elite” DevOps, also known as software developing approach, teams were defined and bench-marked. Many organizations and corporations expected mid-levels to participate in practices other then just code writing like: works on CI/CD pipelines (an automated assembly line for code), cloud deployments, observability, and reliability.[36] Those practices became standard performance targets and expectations, raising the bar for mid-level roles. Overall, as shown in Table 2, the difficulty of the skills required during the Pre-AI period is rated 3 with a bit more span than the entry-level, but still centered on human-written code.

Today, the baseline and basics of coding has not vanished, but applicants are also expected to be AI literate and safe. According to the Stake Overflow survey, 76 percent of developers today use or plan to use AI tools.[31] Hiring managers now look for candidates that can code with assistants effectively as AI becomes a daily tool used in the mainstream. Though the use of AI increases the productivity, problems of reliability arise.[4] Research has shown that AI code suggestions can be insecure which calls for developers to recognize and fix these outputs. Overall, as shown in Table 2, the difficulty of the skills required presently is rated 4 as all you need to understand is the proficiency of handling AI tools.

In the short-term future (2025-2030), the role of computer scientist shifts from good coders to good AI-augmented system builders. Gartner estimates that, a global research and advisory from that provides future insights for various fields, 75 percent of software engineers will use AI assistants by 2028.[37] Mid-level applicants in the future need to grasp an understanding of running AI features safely with LLMOps evaluating the whole generating process paired with retrieval-augmented generation pulling facts from trusted company sources. [38] Know how to apply TEVV (test, evaluation, verification, and validation) and secure-by-design to ensure creditability while managing performance with observability of AI pipeline, the sequence of steps that turns a input to a safe output. Short-term future, with the integration of building and operating AI features, is rated a 5. (Table 2)

In mid-term future, Mid-level developers are turning into AI coordinator who set up systems where different agents and tools work together. As AI shows up in companies, the job shifts from writing every path by hand to designing workflows, integrators, and how tools communicate and connect with each other. Rules are also a huge part of jobs as many different AI regulation and laws are created which leads to actually practice risk checks, documentation, transparency, and incident reporting.[39] For that reason, the difficulty is rated a 5, shown in table 2, as the hard part isn't typing code, but clearly specifying, limiting, and testing what these AI systems can do in a regulated environment.

By 2040 and beyond, mid-level developers will be experts in contract-based development and ongoing assurance. They will turn requirements into formal or semi-formal rubrics that AI systems implement, then verify their behavior through polices, tests, monitoring, similar to mid-term future with a bit more advances. The skills during this period is rated a difficulty of 5 shown in table 2.

In general, while the Pre-AI stage let mid-levels advance through steady feature work and mentorship, the present and future compress this path as AI absorbs repetitive coding and raises expectations for end-to-end delivery and oversight. As AI capabilities expand and early AGI-style autonomy matures, required skills climb sharply from "write code" to "specify, supervise, and validate" AI-enabled systems; accordingly, the ranking of difficulty of required skills in Table 2 increases with time, showing a direct relationship.

5.4.2 Mid-Level Necessity for Up-skilling

Within the Pre-AI era up to the end of 2021, mid level engineers advanced by deepening system design, testing, and delivery using DevOps and SRE playbooks.[36] Observability was a core growth area, which means seeing the real state of a system through logs, metrics, and traces so that issues are found and fixed quickly. The pressure to learn AI specific practices existed only in specialized teams.[31] For this period the urgency to up-skill in AI is moderate and the rating is a 3, as you can see in Table 2.

In the present between the years 2022 and 2025, adoption of AI assistants is broad, and quality control is the deciding skill for mid level roles.[3] Engineers need to pair AI usage with tests, code review, and secure coding so that almost correct suggestions do not enter production.[4] Teams that self train and document how they use AI tend to move faster with fewer errors.[14] For this period the urgency to up-skill is high and the rating is a 4, as you can see in Table 2.

Within the short term future between the years 2025 and 2030, LLMOps becomes routine and that means safe deployment of AI features, task grounded evaluation where tests mirror real work, continuous monitoring, and fast rollback when a result is wrong. Human guidance remains critical because current models still struggle with multi file and multi step changes in large codebases.[20] Compliance expectations expand, so mid level developers will help produce

documentation that explains data sources, model versions, and known risks.[39] For this period the urgency to up-skill is maximal and the rating is a 5, as you can see in Table 2.

Within the mid term future between the years 2030 and 2040, organizations will run full AI management systems with audits, controls, and improvement cycles that work the way quality systems work in other industries.[32] Mid level engineers will grow from implementers into coordinators of systems who connect services, data, and AI features and then prove that these connections are safe and reliable. Career growth will depend on the ability to combine engineering depth with governance and measurement.[6] For this period the urgency to up-skill stays at the maximum and the rating is a 5, as you can see in Table 2.

Within the long term future beyond the year 2040, routine coding will be handled by tools more often, so mid level engineers will spend more time designing workflows, setting guardrails, and proving safety through tests and audits.[6] Incident response, which means the steps taken when something goes wrong, will include AI specific playbooks that roll back changes and notify stakeholders.[38] Data handling and evaluation design will remain human led because these steps require judgment and context. For this period the urgency to up-skill remains at the maximum and the rating is a 5, as you can see in Table 2.

Across the five periods, mid level roles evolve from coding and shipping toward operating and assuring AI enabled systems. The key advantage shifts to those who can join code quality, monitoring, evaluation, and documentation into one reliable workflow. The overall message is that advancement will favor engineers who can prove outcomes under constraints and who can explain how and why a system produced a given result.

5.4.3 Mid-Level Job Market Access

In Pre-AI (Before 2021), Master's graduates enjoyed a skills premium in analytics and ML-adjacent roles as cloud, data, and MLOps stacks expanded across enterprises, which translated into generally moderate access but with pronounced cyclicality tied to funding cycles and macro conditions.[34] In this period, hiring screens emphasized mathematical maturity, experimentation experience, and end-to-end data systems exposure rather than proficiency with AI copilots or prompt-engineering workflows. Employers surveyed for multi-year outlooks anticipated digital transformation and automation, yet did not routinely require generative-AI literacy from Master's-level applicants. This corresponds to an access-difficulty rating of 3; see Table 2.

In the Present (2022–2025), developer and workplace data show widespread AI adoption—76 percent of respondents report using or planning to use AI tools and 62 percent report current use which raises interview expectations for AI-augmented workflows, evaluation, and delivery.[3] In the Kaggle status chart for this cohort, titles labeled "increasing" exceed "decreasing"; Fig. 5), indicating that role variety is expanding for AI-capable Master's candidates who can integrate foundation-model services with conventional software stacks. In parallel, leadership surveys highlight that 75 percent of knowledge workers report using AI at work, reinforcing employer preferences for candidates who can orchestrate tools responsibly and document the resulting workflows. This corresponds to an access-difficulty rating of 3; see Table 2.

In the Short-Term Future (2025–2030), the Kaggle slopegraph shows openings rising by roughly 6.2 percent; Fig. 7), consistent with copilots becoming standard issue across development teams and analytics groups. Enterprises report that AI is moving from individual experimentation to organizational adoption, which pushes interviews toward questions on reliability, evaluation, and safe tool use rather than purely manual coding throughput. Macro analyses suggest that generative AI will contribute to sustained productivity growth through 2040 if redeployment and reskilling are managed effectively, which reinforces demand for Master's graduates who can translate time-savings into higher-value deliverables.[6] This corresponds to an access-difficulty rating of 2; see Table 2.

In the Mid-Term (2030–2040), productivity gains and broader autonomy are expected to concentrate demand on Master's-level talent that can manage agentic systems, data governance, and evaluation pipelines at scale across heterogeneous products.[6] Employer outlooks emphasize reskilling and task reallocation rather than uniform job loss, implying that access remains favorable when AI orchestration and governance capabilities are demonstrable.[34] This corresponds to an access-difficulty rating of 2; see Table 2.

In the Long-Term (2040+), hiring for this cohort is expected to tilt toward AI-enabled product ownership and compliance-aware design, with screening probing systems thinking, safety constraints, and cross-functional delivery histories in addition to fundamentals.[34] Given sustained diffusion of AI throughout enterprise workflows, selection is likely to reward candidates who document operational rigor with models-in-production and post-deployment monitoring. This corresponds to an access-difficulty rating of 2; see Table 2.

In Conclusion , the signal from the Kaggle figures—more titles increasing and openings rising—aligns with adoption and productivity evidence to indicate improving access through 2030 and durable favorability thereafter for AI-literate Master's graduates (Figs. 5, 7). Overall, the access trajectory is upward; see Table 2.

5.4.4 Mid-Level AI's Impact

In Pre-AI (Before 2022), Master's-level tasks centered on analytics backends, modeling experiments, and data engineering patterns, with limited day-to-day dependence on AI copilots or chain-of-tool prompts.[34] Evaluation emphasized reproducibility, documentation, and standard software quality metrics rather than AI-specific practices such as red-teaming or prompt-robustness checks. This corresponds to an AI-impact rating of 2 and an upskilling-urgency rating of 2; see Table 2.

In the Present (2022–2025), controlled experiments and enterprise reports document large time-savings and quality gains when assistants are integrated into common programming workflows, which elevates expectations around throughput and reliability.[4] Developer and workplace surveys confirm mainstream usage of AI tools, making AI-in-the-loop testing and evaluation central to Master's-level practice. The Kaggle impact chart for this cohort skews Moderate/High (Fig. 6), consistent with a shift from manual implementation toward model and service orchestration.[3] This corresponds to an AI-impact rating of 4 and an upskilling-urgency rating of 4; see Table 2.

In the Short-Term Future (2025–2030), work is expected to pivot further toward agent/service orchestration, dataset curation, and safety-aware integration as copilots become default across the stack, strengthening the premium on systems thinking and traceable evaluation.[6] Hiring signals emphasize ownership of end-to-end reliability and cost/performance trade-offs under AI assistance, reinforcing a migration from "writing every line" to "designing and supervising the system." This corresponds to an AI-impact rating of 5 and an upskilling-urgency rating of 5; see Table 2.

In the Mid-Term (2030–2040), increasing autonomy implies that human effort concentrates on architecture, constraints, and human-factors evaluation rather than routine coding, with escalation paths for failure-mode audits and incident response becoming part of standard responsibilities.[34] As consequence, performance reviews at this level are likely to weigh governance artifacts and continuous validation alongside delivery metrics. This corresponds to an AI-impact rating of 5 and an upskilling-urgency rating of 5; see Table 2.

In the Long-Term (2040+), AI pervades product lifecycles, concentrating human value in stewardship, cross-domain design, and accountability for socio-technical outcomes, with emphasis on monitoring drift, bias, and misuse at scale.[34] Given persistent diffusion across industries, continuous learning becomes a standing requirement as practices and guardrails evolve. This corresponds to an AI-impact rating of 5 and an upskilling-urgency rating of 5; see Table 2.

In Conclusion (Impact), experimental and enterprise evidence, together with the Kaggle impact distribution, indicate a strong upward shift toward orchestration, governance, and evaluation over hand coding for Master's-level contributors (Fig. 6). Overall, the impact trajectory is a strong incline; see Table 2.

5.5 PhD Students / AI Researchers

This group is filled with people who create new methods or study advanced systems in AI and computer science. PhD students, academic or industrial researchers, and research engineers who turn ideas into working prototypes are all types of individuals in this group. Their role is to design experiments, manage data, train and evaluate models, and communicate results through papers,

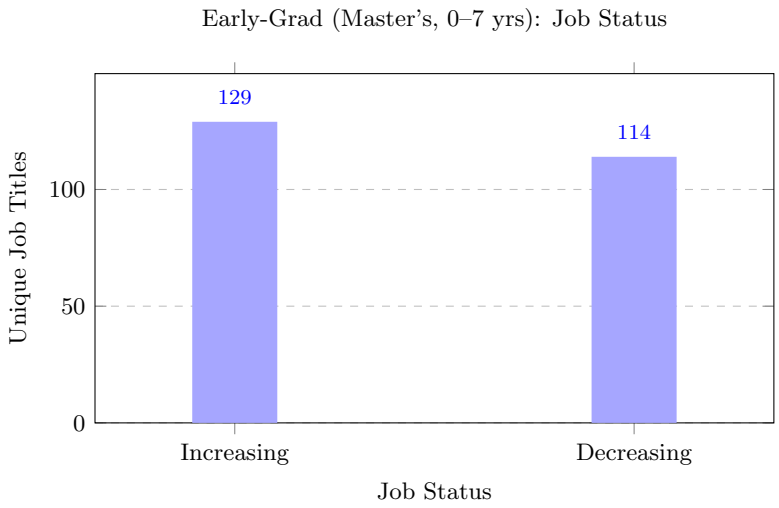


Figure 5: Master's — unique job titles labeled Increasing vs Decreasing.

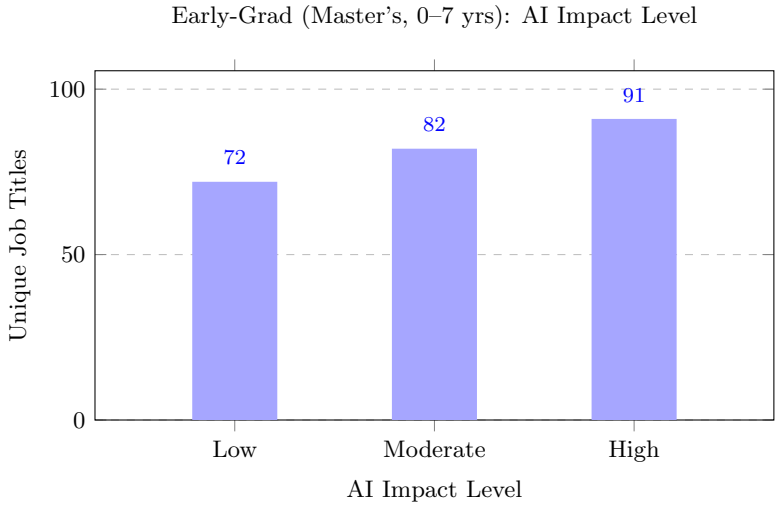


Figure 6: Master's — unique job titles by AI impact level.

Table 2: Mid-Level Coders: Workforce Impact Evaluation Scale

Dimension	Pre-AI	Present	Future Short-term	Future Mid-term	Future Long-term
Required Skills	3	4	5	5	5
Job Market Access	4	3	3	2	2
AI's Impact	2	3	4	5	5
Necessity for Up-skilling	2	4	4	5	5

Note: Pre-AI = before 2022; Present = 2022–2025; Future Short-term = 2025–2030; Future Mid-term = 2030–2040; Future Long-term = 2040 and beyond.

reports, or demos. They are often stereotyped to be "math-only," "paper-first," or disconnected from real-world products. Competition is very high due to fast publication cycles, limited data access, and limited conference slots. Limits include a high bar for originality, reproducibility requirements,funding restrictions, and the challenge of turning research into real value.

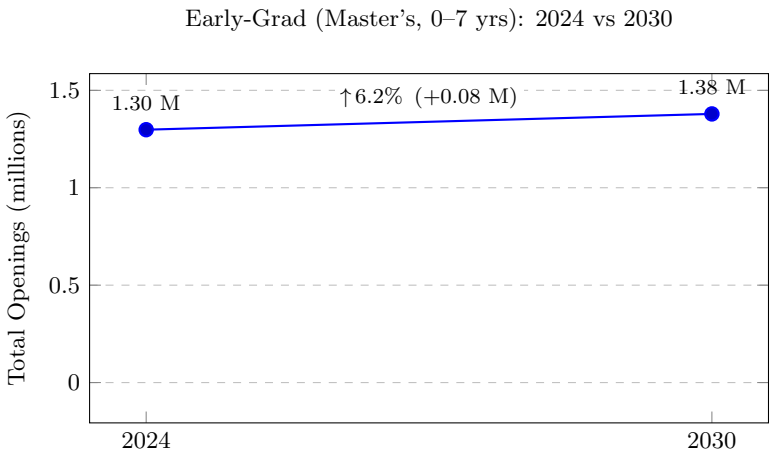


Figure 7: Master's — projected openings increase by about six percent.

5.5.1 PhD Students / AI Researchers Required Skills

Before 2022, AI researchers needed strong math and statistics knowledge, deep learning frameworks like PyTorch or TensorFlow, and solid experimental design mindset and skills. [40] Reproducibility, clear, and general reporting were the core expectations as different publication institutes pushed authors to release code, specify the computing process, and document a well methodology. [7] Researchers also adopted systematic documentations for datasets and models, while recognizing the hidden maintenance costs of ML pipelines, a step-by-step process that turns working machine learning systems in production.Together, these standards raised the bar beyond pure modeling with a rating of 4 shown in table 3.

With the rise of foundational models, employers present-day expect PhD-level hires to master LLM evaluation like: retrieval-augmented generation (RAG), which is a model that looks up relevant documents online in a timely manner, and parameter-efficient tuning, the training of smaller add-on layers instead of the whole model itself, to run fast and affordable experiments. [7] Equally important to that are risk and safety skills as teams and large corporate who look for people who can align with the NIST AI Risk Management Framework and its 2024 Generative AI Profile. Job market surveys show hiring managers now prefer candidates with hands-on AI tool skills, even over those with more then years of experience.[38] In summary receiving a rating of 5, shown in the table 3, as researchers are hired not only to invent models, but to evaluate, adapt, and safely operate them.

In short-term future from 2025 to 2030, required skills are estimated to move from modeling and designing to systems and operations like: building LLM workflows, LLMOps, and task-grounded evaluation on realistic suites that tie research to a products impact. [41] Regulatory literacy becomes a baseline meaning that researchers will be expected to produce documentation, testing, and risk evidence that follows the laws and policies. An expected growing demand for people who can reduce, customize, and launch models under execute and cost limits while maintaining safety controls. These advanced capabilities cause the the difficulty of these skills to be ranked a 5 on the table 3 which also make these researchers directly valuable to product teams.

By the 2030s, industry labs will lean on researchers who can co-design science and AI, workflows you can rerun the same way across different data types, and review outputs in large amounts. Macro reports point to fast skill churn and rising demand for AI methods + governance literacy, not just model building. [42] AI remains a primary driver of job change, so researchers who can mix technical depth with evaluation, policy awareness, and cross-field collaborations will stand out. The role blends research, engineering, and responsible operations rather than pure algorithm design which also ranks the over required skills a 5 on table 3.

Past 2040, many organizations will treat AI like other critical infrastructure, so researchers will need to coordinate and check increasingly autonomous systems and prove they meet organiza-

tions wide AI management standards.[32] Expect formal compliance and continuous monitoring to be everyday skills, supported by certifiable frameworks. That means writing evaluations and reports that satisfy both technical and governance requirements, not just papers and benchmarks. Deep technical range plus operational rigor becomes the default expectations which makes the difficulty ranking a 5 (see table 3).

In general, Pre-AI research centered on modeling and careful experiments, but the present and future add stricter evaluation, data practices, and system-level integration—making the role more demanding. As AI's abilities grow and AGI foundations mature, required skills shift toward originality, rigorous validation, and safe deployment; thus, the ranking of difficulty of required skills in Table 3 rises across the periods in a direct relationship with time.

5.5.2 PhD Students / AI Researchers Necessity for Up-skilling

Within the Pre- AI era up to the end of 2021, research up-skilling emphasized reproducible pipelines, which means steps that can be repeated by others with the same results, and awareness of hidden technical debt in machine learning systems, which means long term costs that arise from quick fixes.[43] The community promoted datasheets for datasets, which are short standardized records that describe how data was created, what it contains, and how it should be used.[44] These practices built a base for responsible studies and transparent publication. For this period the urgency to up-skill is moderate and the rating is a 3, as you can see in Table 3.

In the present between the years 2022 and 2025, generative AI accelerates drafting and coding, which increases the need for low cost evaluation and careful human checking to avoid weak claims and wrong results. Field studies show real productivity gains but also show that structure, metrics, and guardrails are required to keep quality high.[14] Benchmarks still reveal gaps on complex code changes, which means human expertise and review remain essential.[20] For this period the urgency to up-skill is high and the rating is a 4, as you can see in Table 3.

Within the short term future between the years 2025 and 2030, task grounded evaluation becomes a core skill, which means testing models on tasks that match real scientific or engineering work.[**?**] Data governance also becomes central, which means clear rules for how data is collected, labeled, stored, and used, and clear records of approvals and risks. Researchers who can link evaluation to real impact will publish stronger papers and ship safer prototypes. For this period the urgency to up-skill is maximal and the rating is a 5, as you can see in Table 3.

Within the mid term future between the years 2030 and 2040, standards and laws will require clearer documentation, stronger risk controls, and better auditability of research code and data.[39] Auditability means the ability to trace and explain how a result was produced so that any reviewer can check it step by step. As more routine coding and drafting is automated, originality and value will come from problem framing, study design, and careful evaluation methods.[6] For this period the urgency to up-skill stays at the maximum and the rating is a 5, as you can see in Table 3.

Within the long term future beyond the year 2040, AI agents will likely handle more of the research pipeline such as running experiments and drafting results, which pushes humans to lead on question choice, constraint setting, ethics review, and stronger benchmarks that are harder to game.[20] Evaluation design will remain a human strength because it requires judgment about what truly counts as progress.[6] Governance literacy will also be necessary so that research can be audited and trusted.[38] For this period the urgency to up–skill remains at the maximum and the rating is a 5, as you can see in Table 3.

Generalization for PhD students and researchers Across the five periods, the center of effort moves from writing code and models to designing sound evaluations, curating trustworthy data, and explaining results in a way that stands up to review. The competitive edge comes from the ability to connect methods to measurable outcomes while keeping risks controlled and traceable. The overall message is that research careers will favor rigorous evaluators and careful stewards of data and systems, not only fast coders.

5.5.3 PhD Students / AI Researchers Job Market Access

In Pre-AI (Before 2022), doctoral candidates were hired into academic labs, industrial research, and specialized engineering roles where modeling, inference, and experimental design were central, yielding moderate access with strong competition for elite posts.[34] Hiring signals stressed publication quality and research rigor more than AI-tool operation per section, reflecting pre-generative-AI evaluation norms. This corresponds to an access-difficulty rating of 3; see Table 3.

In the Present (2022–2025), the Kaggle status chart shows more titles increasing than decreasing; Fig. 8) and openings that trend upward toward 2030, consistent with healthy access for adaptable researchers who bridge research and governed deployment. Employer outlooks report rising demand for applied research that integrates LLMs into production systems with appropriate safeguards, expanding translational roles at the research–product interface.[34] This corresponds to an access-difficulty rating of 2; see Table 3.

In the Short-Term Future (2025–2030), the Kaggle slopegraph indicates openings increasing by 6.1 percent ; Fig. 10), suggesting more requisitions for researchers who can deliver publication-plus-product impact and data stewardship. Industry trend reports anticipate that organizations will continue to invest in safety reviews, evaluation suites, and post-deployment monitoring, creating demand for doctoral expertise in methods and measurement.[34] This corresponds to an access-difficulty rating of 2; see Table 3.

In the Mid-Term (2030–2040), as assistants assume routine research coding, hiring preferences are expected to emphasize experiment design, multi-modal data management, and robust evaluation practices, including causality-aware analyses and stress-testing.[34] Macro projections of continuing productivity gains reinforce institutional incentives to recruit researchers who translate foundational advances into reliable, human-centered systems.[6] This corresponds to an access-difficulty rating of 2; see Table 3.

In the Long-Term (2040+), access remains favorable but competition becomes more global and specialization-driven, with premiums on alignment, verification, and domain science for research-intensive product organizations.[34] As scientific workflows scale with AI agents, hiring is expected to prioritize stewardship capabilities that ensure reproducibility and external validity. This corresponds to an access-difficulty rating of 2; see Table 3.

In Conclusion, the Kaggle tilt toward increasing titles and the upward openings trend are consistent with employer and macro evidence for improving and durable access for PhD talent that translates research into governed deployment (Figs. 8, 10). Overall, the access trajectory is upward; see Table 3.

5.5.4 PhD Students / AI Researchers AI's Impact

n Pre-AI (Before 2021), tooling assisted research workflows but did not materially reshape core processes for most early-career researchers, leaving publication cadence, experimental rigor, and collaboration as the main evaluation axes.[34] This corresponds to an AI-impact rating of 2 and an upskilling-urgency rating of 2; see Table 3.

In the Present (2022–2025), controlled and field results document accelerations in literature review, coding, and experiment setup with assistants, while developer surveys confirm mainstream AI usage in technical work.[4]Increased reliance on AI-mediated drafting and analysis elevates the importance of traceability, error analysis, and clear attribution in research artifacts.[34] This corresponds to an AI-impact rating of 4 and an upskilling-urgency rating of 4; see Table 3.

In the Short-Term Future (2025–2030), agentic workflows are expected to streamline data preparation, evaluation, and drafting, shifting human differentiation toward originality, causal identification, and safe deployment under constraints.[34] Organizations are likely to value hybrid profiles that combine methodological rigor with productized evaluation and monitoring experience. This corresponds to an AI-impact rating of 5 and an upskilling-urgency rating of 5; see Table 3.

In the Mid-Term (2030–2040), research emphasis moves further toward governance and real-world impact over synthetic benchmarks, with reproducibility, robustness, and external validity becoming central to funding and promotion decisions.[34] Persistent productivity improvements strengthen the case for researchers who can quantify and mitigate failure modes in deployment

contexts.[6] This corresponds to an AI-impact rating of 5 and an upskilling-urgency rating of 5; see Table 3.

In the Long-Term (2040+), AI saturation in R and D positions scientists to curate hypotheses and causal tests while systems perform large-scale execution, multi-modal synthesis, and rapid iteration across domains.[34] Research evaluation is expected to weigh stewardship of powerful tools and social impact alongside novelty metrics. This corresponds to an AI-impact rating of 5 and an upskilling-urgency rating of 5; see Table 3.

In Conclusion, doctoral-level work is transformed toward design, safety, and causal inference as assistants absorb more routine experimental tasks, which aligns with the rising emphasis on governance and impact in research practice (Fig. 9). Overall, the impact trajectory is a strong incline; see Table 3.

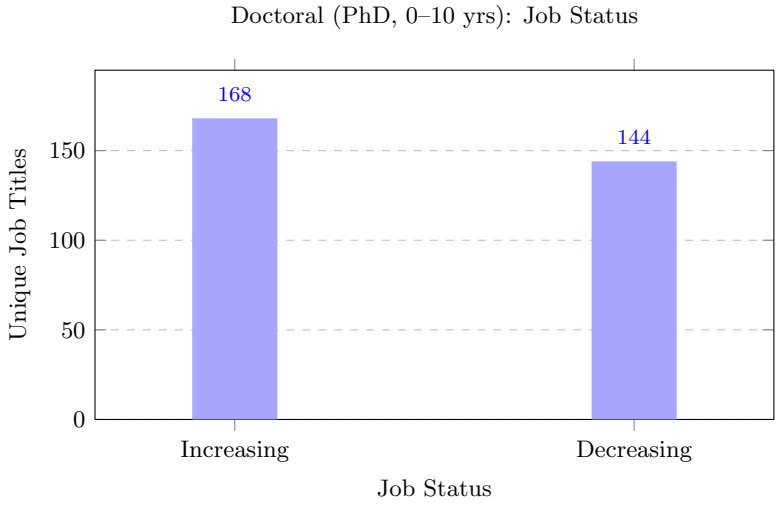


Figure 8: PhD — unique job titles labeled Increasing vs Decreasing.

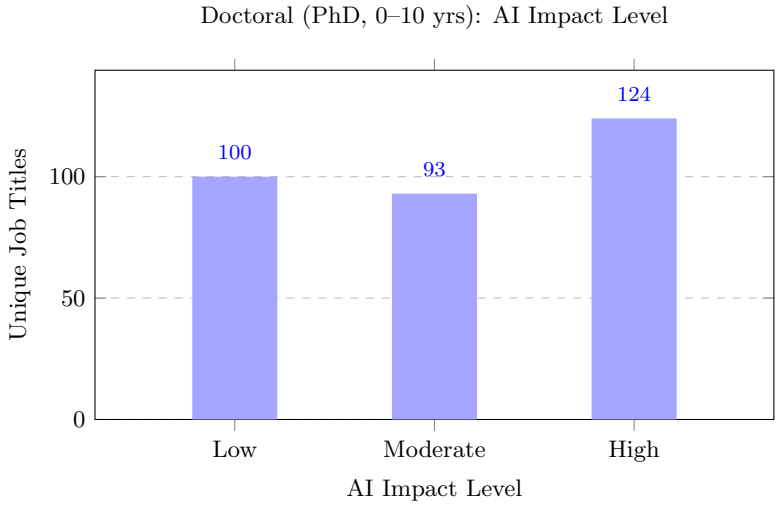


Figure 9: PhD — unique job titles by AI impact level.

5.6 Long-Time Professionals / Managers

In this paper, long-time professionals, known as seniors or managers, are engineers and IT individuals with around 10+ years of experience that are reliable, secure, and collaborative on

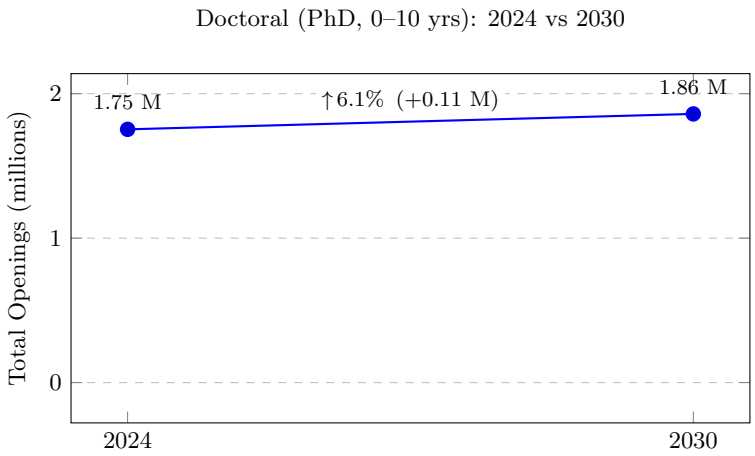


Figure 10: PhD — projected openings increase by about six percent.

Table 3: PhD Students / AI Researchers: Workforce Impact Evaluation Scale

Dimension	Pre-AI	Present	Future Short-term	Future Mid-term	Future Long-term
Required Skills	4	5	5	5	5
Job Market Access	4	4	3	3	2
AI's Impact	2	3	4	5	5
Necessity for Up-skilling	2	3	4	5	5

Note: Pre-AI = before 2022; Present = 2022–2025; Future Short-term = 2025–2030; Future Mid-term = 2030–2040; Future Long-term = 2040 and beyond.

different projects. Their role is to make big technical decisions, mentor others, coordinate with products design, and balance different trade-offs, like cost vs. performance vs. safety. Many stereotypes of them include "more paperwork than problem-solving" and "rely on older tools, avoiding new ones". Competition comes from smaller teams using AI to move faster and from the younger, lower-cost talent. Key limits include keeping skills up-to-date during rapid technical shifts, managing risk in large legacy systems, and navigating layoffs or reorganizations tied to the effects of AI automation.

5.6.1 Long-Time Professionals / Managers Required Skills

Senior engineers and engineering managers were expected to master software, managements, and people who own leadership grounded within the Software Engineering Body of Knowledge, also known as SWEBOK.[45] They had operational excellence centered on DevOps, coordination between development and IT operations to ship software quickly and safely, and CI/CD, automated build-test-deploy pipelines, with leaders implementing practices that correlate with top performances.[46] They are excellent leaders who are critical to monitoring, assigning tasks, leading groups, and responding to incidents so that teams can manage outages and recover in a timely manner. As shown in table 4, the difficulty of the required skills is rated a 4.

AI coding tools are now implemented into the mainstream, so leaders must adapt and organize for require human review of AI output and set clear AI-use rules and guidelines in teams. These leaders are required to run AI governance and ensure that AI is managed properly, so that outputs and systems used are secure, creditable, and safe. [47]Though studies show that AI automation can increase productivity by 56 percent, many research also show that AI code may be insecure and unreliable, managers have to guide teams to find the right balance between the two.[4] As shown in table 4, the difficulty of the required skills is rated a 4.

Within the short-term future, seniors are still required to have leadership skills and basic soft skills, however, they also have to change and prepare for LLMOps. Multiple employer surveys show that the skill mix is changing fast through 2030, with leadership, analytical thinking, and

AI/big-data literacy moving up the priority list—so senior professionals will spend more time orchestrating people+AI workflows and less time on manual coding. Lastly, The most important, core skill, is setting the strategy and measurements of what, where, and why use AI. As shown in table 4, the difficulty of the required skills is rated a 5.

During mid-term future, leaders will manage agentic AI—software "agents" that can take multi-step actions—by specifying guardrails and accountability, which shifts skills toward workflow design, monitoring, and risk management rather than hand-coding every path.[48] Research roadmaps for software engineering to 2030 highlight sustained needs in verification and validation, human-centered engineering, and AI-for-SE—in plain terms, managers must prove systems behave as intended and keep humans in the loop.[26] Because employers expect a large share of skills to change by 2030, mid-career pros and managers will lean on continuous reskilling and talent management (coaching teams to use AI well) as durable, high-value competencies rating the required skills a 5 shown on the table 4.

If today's trends continue (inference based on current evidence), senior roles will look like "specify, govern, and assure": managers will write precise requirements for AI systems, run assurance programs (evidence that systems are safe, secure, fair), and maintain organization-wide AI management systems rather than expanding headcount for routine coding.[26] Given forecasts that leadership, social influence, and analytical thinking keep rising in importance, the durable edge for long-time professionals will be human judgment under uncertainty—making strategic calls, navigating regulation, and aligning AI to business and ethics. As shown in table 4, the difficulty of the required skills is rated a 5.

In general, the Pre-AI era already required high standards in architecture, reliability, and team leadership, but the present and future add AI governance (policies and controls for safe and fair AI use), regulatory alignment, and oversight of autonomous agents. As AI advances and AGI foundations mature, required skills broaden and deepen toward specifying, governing, and assuring AI systems at scale; consequently, the ranking of difficulty of required skills in Table 4 increases over time and remains at the high end.

5.6.2 Long-Time Professionals / Managers Necessity for Up-skilling

Within the Pre- AI era up to the end of 2021, senior up-skilling centered on leading reliable systems, mentoring teams, and improving delivery using SRE and DevOps practices.[36] Observability and incident response were core leadership duties, where incident response means the steps used to diagnose and fix problems in production systems. AI topics were active in niche teams but were not central in most senior job descriptions. For this period the urgency to upskill in AI specific areas is moderate and the rating is a 3, as you can see in Table 4.

In the present between the years 2022 and 2025, organizations are adopting AI at speed, so senior roles include setting policy, selecting tools, and managing risk along with architecture and coding. Hiring and workload mixes are changing, which raises the need to reskill teams and to define safe ways of working with AI.[6] Surveys also show trust gaps, which keeps governance and review on the leadership agenda.[31] For this period the urgency to up-skill is high and the rating is a 4, as you can see in Table 4.

Within the short term future between the years 2025 and 2030, AI coding tools become standard and leaders must guide AI strategy, evaluation at scale, and guardrails for privacy and safety. Guardrails are rules and checks that prevent unsafe or insecure behavior in systems that use AI.[38] Compliance work expands as well, which means leaders need literacy in documentation, risk classes, and incident reporting, and they must train their teams to meet these expectations.[39] For this period the urgency to up-skill is maximal and the rating is a 5, as you can see in Table 4.

Within the mid term future between the years 2030 and 2040, AI management systems will be as normal as security and quality systems are today, and senior engineers will run audits, controls, and continuous improvement cycles.[32] The biggest gains from AI will come when leaders redesign roles and processes so that human strengths and AI strengths complement each other.[6] This requires ongoing learning in both technology and organizational change. For this period the urgency to up-skill stays at the maximum and the rating is a 5, as you can see in Table 4.

Within the long term future beyond the year 2040, as AI handles more routine work, senior roles will center on architecture, policy, and human AI teamwork.[6] Leaders will be measured by safe outcomes and business impact rather than by lines of code, which means steady learning in evaluation, risk management, and communication. The ability to explain decisions, to show evidence, and to correct course quickly will define excellence.[38] For this period the urgency to up-skill remains at the maximum and the rating is a 5, as you can see in Table 4.

Generalization for long time professionals and senior engineers Across the five periods, leadership expands from reliability and delivery toward stewardship of AI enabled systems. The most valuable leaders will be those who can connect architecture, safety, compliance, and team development into a single plan and who can show proof that the plan works in production. The overall message is that senior careers will reward continuous learning in strategy, assurance, and people development as much as raw technical output.

5.6.3 Long-Time Professionals / Managers Job Market Access

In Pre-AI (Before 2022), senior access was broadly solid because experience signaled delivery, architecture, and leadership in complex systems, and demand was supported by large-scale modernization and cloud migrations.[34] Screening emphasized design trade-offs, incident histories, and mentorship rather than AI-specific practices. This corresponds to an access-difficulty rating of 2; see Table 4.

In the Present (2022–2025), the Kaggle figures are near-balanced with a slight tilt to "increasing"; Fig. 11) and openings up 2.4 percent by 2030 (Fig. 13), pointing to reasonable access when leaders can steer AI-augmented teams and manage risk. Employer evidence also emphasizes organization-wide adoption challenges that reward senior candidates who can codify responsible-use patterns and operating procedures.[10] This corresponds to an access-difficulty rating of 3; see Table 4.

In the Short-Term Future (2025–2030), as AI reduces repetitive coding headcount, access concentrates in leadership, formalization, and governance roles best served by experienced engineers able to institutionalize AI practices.[10] Enterprise materials highlight the premium on leaders who can measure impact and reliability under mixed human-and-AI delivery, reinforcing demand at the senior tier. This corresponds to an access-difficulty rating of 3; see Table 4.

In the Mid-Term (2030–2040), demand is expected to favor senior talent who manage agentic systems, SLAs, and cross-team accountability, including cost controls, audit trails, and safety reviews embedded in pipelines.[34] Macro projections of persistent productivity gains support continued organizational investment in orchestration and governance capacity at senior levels.[16] This corresponds to an access-difficulty rating of 3; see Table 4.

In the Long-Term (2040+), access should stabilize around fewer but broader roles centered on orchestration, governance, and policy compliance as AI pervades delivery, with portfolio breadth and accountability histories becoming decisive.[34] Large-scale adoption dynamics suggest durable need for leaders who can align socio-technical risk with business goals. This corresponds to an access-difficulty rating of 3; see Table 4.

In Conclusion, the near-balanced status chart and modest openings increase in the Kaggle figures are consistent with flat-to-slightly-tighter access concentrated in leadership and architecture (Figs. 11, 13). Overall, the access trajectory is stable to mildly down; see Table 4.

5.6.4 Long-Time Professionals / Managers AI's Impact

In Pre-AI (Before 2021), senior engineers automated via scripts and CI/CD, but AI's role in day-to-day delivery remained limited compared with current practice, keeping impact primarily in the realm of conventional tooling and process improvement.[34] This corresponds to an AI-impact rating of 2 and an upskilling-urgency rating of 2; see Table 4.

In the Present (2022–2025), Copilot-style tools accelerate refactors and code reviews, with randomized evidence of -55 percent faster task completion and field reports of quality benefits under AI assistance, reshaping expectations around throughput and design diligence.[4] Workplace studies indicate that use of AI at work has nearly doubled, reaching -75 percent of knowledge workers, which pushes senior responsibilities toward setting policy, reviewing prompts, and

mentoring AI-augmented teams. This corresponds to an AI-impact rating of 3 and an upskilling-urgency rating of 4; see Table 4.

In the Short-Term Future (2025–2030), leaders are evaluated on orchestration and responsible use as AI manages more backlog, integration, and observability across services, making cost, reliability, and safety metrics part of routine performance discussions.[16] Enterprise guidance emphasizes measuring developer experience and downstream outcomes under Copilot-enabled workflows, reinforcing the managerial dimension of senior roles. This corresponds to an AI-impact rating of 4 and an upskilling-urgency rating of 4; see Table 4.

In the Mid-Term (2030–2040), agentic AI absorbs more ops/test work, shifting human focus to constraints, reliability, cross-team risk management, and auditability, especially for multi-service programs with external regulatory interfaces.[34]Persistent productivity gains strengthen the case for senior engineers who can operationalize governance without throttling delivery.[16] This corresponds to an AI-impact rating of 4 and an upskilling-urgency rating of 5; see Table 4.

In the Long-Term (2040+), human value concentrates on system intent, governance, and cross-functional accountability as AI saturates delivery and maintenance, requiring senior leaders to arbitrate trade-offs among safety, privacy, cost, and speed.[34] Enterprise adoption narratives suggest that continuous learning, red-teaming practice, and policy refinement become permanent features of senior roles. This corresponds to an AI-impact rating of 4 and an upskilling-urgency rating of 5; see Table 4.

In Conclusion, scope moves from direct implementation toward orchestration and accountability, with pervasive productivity assistance altering expectations for mentorship, design review, and incident response in AI-heavy environments. Overall, the impact trajectory is a moderate incline; see Table 4.

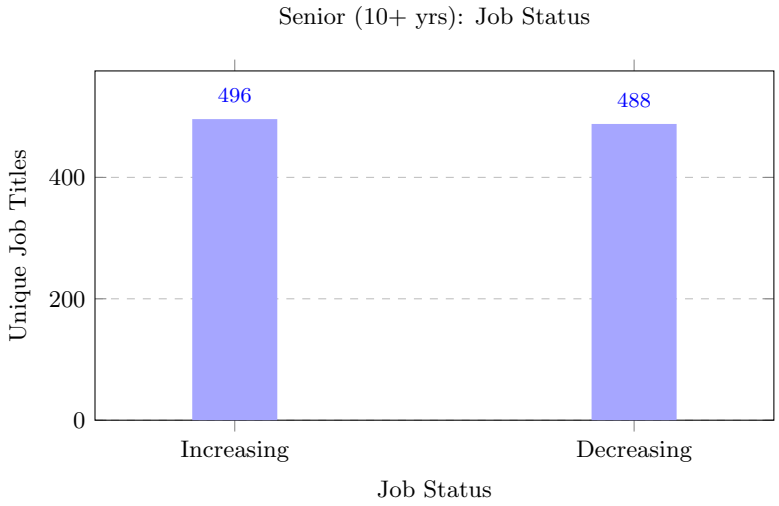


Figure 11: Senior — unique job titles labeled Increasing vs Decreasing.

Table 4: Senior Engineers / Long-Time Professionals: Workforce Impact Evaluation Scale

Dimension	Pre-AI	Present	Future Short-term	Future Mid-term	Future Long-term
Required Skills	4	4	5	5	5
Job Market Access	4	3	3	2	2
AI's Impact	1	3	4	5	5
Necessity for Up-skilling	1	4	4	5	5

Note: Pre-AI = before 2022; Present = 2022–2025; Future Short-term = 2025–2030; Future Mid-term = 2030–2040; Future Long-term = 2040 and beyond.

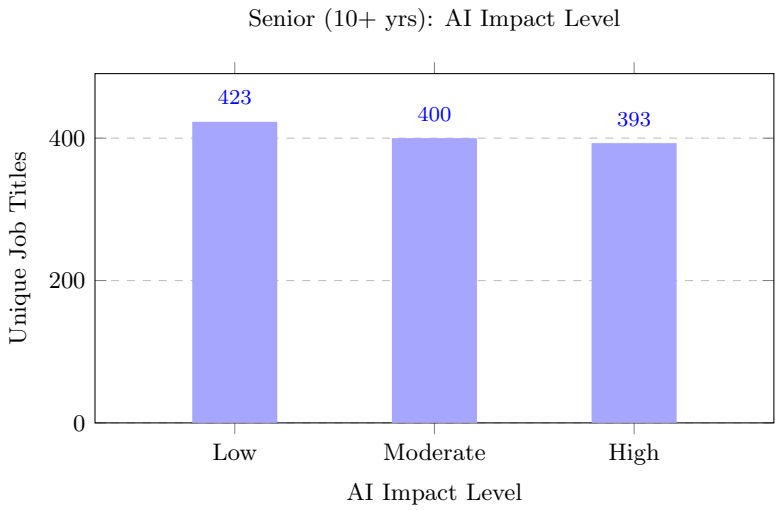


Figure 12: Senior — unique job titles by AI impact level.

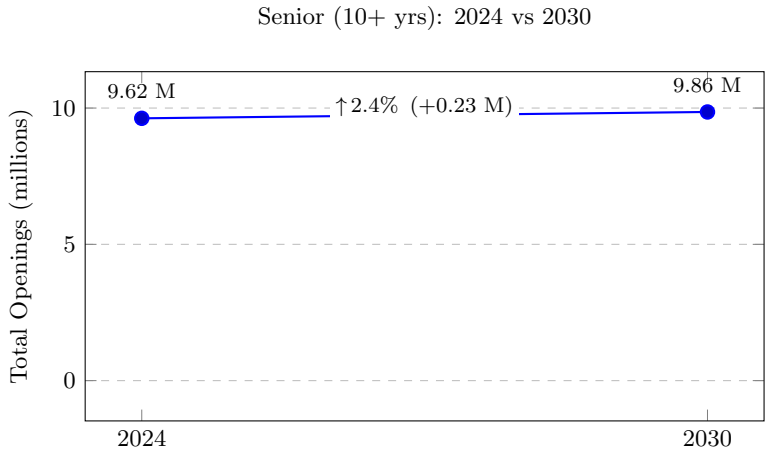


Figure 13: Senior — projected openings increase modestly.

6 Compare and Contrasting the Different Field Together

This section integrates the groups analyses in Section 5 together finding intersecting patterns. To enable a single-view comparison, a composite "workforce readiness index" is defined as the mean of the four Section-5 dimensions on the 1–5 scale. Higher values indicate a more demanding environment. Figure 14 visualizes the trajectories for all four groups across the five time periods; the underlying groups ratings are taken directly from Tables 1–4.

6.1 Values AI is Changing

Figure 14 shows an upward trajectory from Pre-AI to 2030–2040+ for all groups, indicating a shift from manual coding toward specification, orchestration, verification, and governance. In Pre-AI, curves sit in the mid-range, consistent with the "Required Skills" rows in Tables 1–4. In the Present (2022–2025), each curve rises as AI literacy becomes baseline; "AI's Impact" and "Up-skilling" increase to 3–4 across groupss in Tables 1–4, which appears as a clear step up in Figure 14. The Short-Term Future (2025–2030) brings another lift as teams run AI features in production; "Required Skills" reaches 4–5 and "AI's Impact" moves toward 4–5, matching the steeper mid-section of the lines. By 2030–2040+, the four curves converge near the top band, mirroring the 4–5 values across the "Required Skills" and "Necessity for Up-skilling" rows of

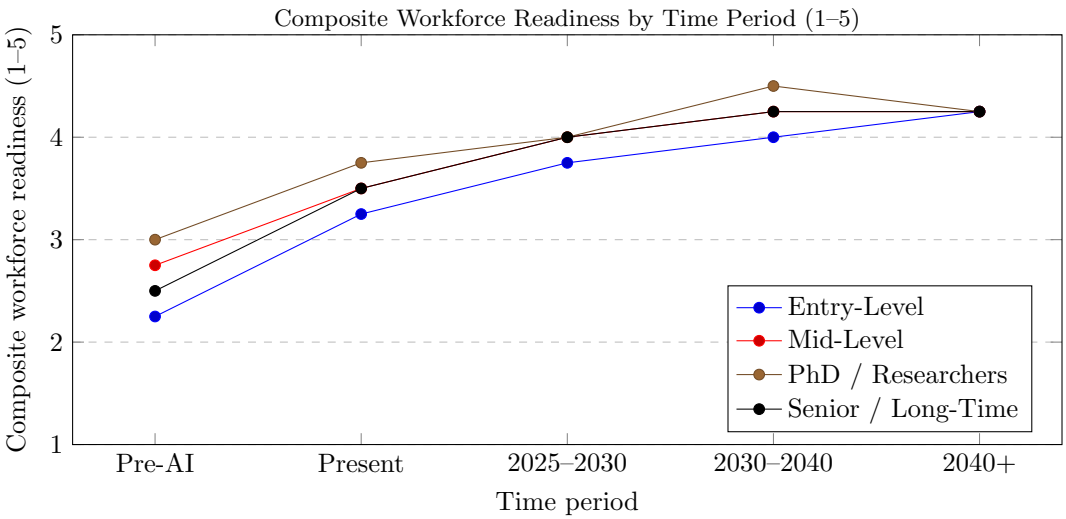


Figure 14: Comparing composite of workforce readiness. The composite equals the mean of four Section 5 dimensions (Required Skills, Job Market Access, AI's Impact, Necessity for Up-skilling) for each group and period. Higher values indicate a more demanding environment.

Tables 1–4.

6.2 New Roles Created

Role evolution follows the shapes in Figure 14. Entry-level contributors move toward AI-aware QA and safety stewardship as "AI's Impact" and "Up-skilling" climb from low Pre-AI values to the top of the scale (Table 1). Mid-level developers shift into LLMOps and agent-workflow integration, managing guardrails, observability, and cost–reliability trade-offs; sustained 5s in "Required Skills" and "Up-skilling" in Table 2 align with the higher mid-period composite. PhD students and research engineers specialize in evaluation science and data curation linked to governed deployment; the prominent mid-term rise in the PhD line corresponds to strong "Required Skills" and high "AI's Impact" in Table 3. Senior professionals consolidate into AI product ownership and governance leadership; high values in "Required Skills," "Up-skilling," and "AI's Impact" in Table 4 keep the senior curve near the top of Figure 14.

6.3 Labor Polarization

Polarization appears in Figure 14 as temporarily wider gaps among groups in the Present and Short-Term Future, followed by convergence at high demand by 2030–2040+. Section-5 evidence indicates a near-term squeeze at the entry tier, then stabilization once AI-first pathways standardize. The dip in "job market access" for Entry in Table 1 aligns with the later catch-up of the entry curve shown in figure 14. Mid-level and long-term professions access improves after AI-operations capabilities are demonstrated, reflected by the "job market access" rows in Tables 2 and 4 and by relatively higher mid-term composites. PhD and researchers roles are seen strong mid-term demand for evaluation, data governance, and translational impact. The PhD curve's rise in Figure 14 mirroring sustained 5s in "required skills" and elevated "AI's Impact" in Table 3. Overall, polarization is driven less by absolute job loss and more by a premium on documented orchestration and assurance skills, which the figure visualizes as higher composites for groups that demonstrate these capabilities sooner.

7 Discussions and Reflection

7.1 Views

In my view, AI is not eliminating software work so much as redefining it. The most valuable contributions now come from specifying problems clearly, integrating AI tools safely, and validating results end-to-end, not from typing the most lines of code. This is consistent with Figure 14, where all workforce groups trending toward a high-demand environment by 2030–2040 and beyond, and with recent developer surveys showing broad AI adoption in day-to-day workflows.

AI raises the bar for beginners; however, it also opens paths for those who can orchestrate AI with tests, data governance, and monitoring. Industry reports word this as moving from "experimentation" to business transformation, which matches the rise in "Required Skills" and "Up-skilling" across Tables 1–4. Finally, trust and safety will be remained to be lead by humans for security.

7.2 Influence on Career Pathway

These findings reshape plans in concrete ways. First, coursework will emphasize on foundations and operations like: linear algebra, probability/statistics, algorithms, data systems, and software testing paired with ethics and policy. Second, assets of work will prioritize accountability so showcasing 2–3 public repos with prompts, retrieval configurations, evaluation harnesses, and unit tests so that reviewers can see how AI output was verified. Third, internships and research rotations become essential to learn LLMOps and observability in real systems; even as conversion rates change, internships remain a primary roles and practical experience.

Finally, daily development will incorporate AI responsibly (pairing generation with tests and reviews) to capture productivity gains without sacrificing reliability. Evidence from Copilot studies and enterprise reports suggests material time savings when teams redesign workflows around AI.

7.3 Recommendations for The Workforce

The recommendations below reflect the comparison trends in Figure 14 and the dimension scores in Tables 1–4.

7.3.1 Suggestions for Entry-Level

Entry-level candidates benefit from the basic, AI-aware. Core programming, data structures, testing, and version control, then with the layers in with. Publishing 2–3 small GitHub repositories are very beneficial including: show the prompts and settings used, include unit tests that catch model errors, and contain a short model card stating limits and fixes. Pursuing internships or open-source contributions to gain early production habits and mentorship still remain a key pathway to offers even as employer conversion rates vary year to year.

7.3.2 Suggestions for Mid-Level

Mid-level developers should own features end-to-end with AI in the loop: requirements, prompt/ workflow design, retrieval/data controls, tests/monitors, releasing and post-incident learning. Track lead time for PRs, defect rates, and reliability to demonstrate that AI increased throughput without eroding quality. Align day-to-day practices with recognized guidance such as NIST AI RMF and ISO/IEC 42001 so that work remains verifiable.

7.3.3 Suggestions for PhD and Researchers

Pairing original contributions with standardized evaluation like: Release code, data, and evaluation scripts, compare against strong baselines on realistic suites, state failure modes and limits explicitly. When capable, include data notes and a short risk or impact section inspired by

governance frameworks to improve downstream reuse and industry translation. Where AI tools assisted writing or coding documents are checked by humans.

7.3.4 Suggestions for Long-Time Professionals

Long-term professionals and IT leaders should select fit-for-purpose tools, set clear operating rules, and guide the teams so AI is a copilot, not an autopilot. Establishing measurable outcomes like: cycle time, escaped defects, reliability SLOs, and cost-to-serve, so that AI adoption measurably improves delivery. Team should practice to organization-level standards and to organize plans that reward up-skilling. Industry studies show that organizations who move beyond experimentation and formalize these practices are more likely to see durable productivity gains.

8 Future Works

8.1 Building Better Datasets

In the future, predictions should move beyond a single community Kaggle file toward a multi-source AI workforce panel that merges: official labor statistics; large-scale job postings and employer. The inputs for data generation should share a common structured role, sector, and geography, with language detection, outlier checks, and a small expert-labeled set to quantify tagging error. This will ensure for datasets formed to be more reliable for use and estimations for future impacts or trends. Basic privacy and research controls should also be built in.

8.2 Future Analysis

In the future, sectioned revisits should be made to compare the predicted with the future present. With each review, a refresh sources and regenerate of the four Section-5 dimensions should be made to recompute the composite figure. Compare the predictions with real outcomes helps determine the accuracy and trustablilty for future analysis and datasets. Treating the study as a living measurement program preserves comparability and sharpens claims about near-term polarization and long-run changes visible in the mixed components and figures.

9 Conclusion

With the comparison of all the analysis, there is a visible consistent rise across the different groups, indicating a structural shift from manual code production toward specification, orchestration, verification, and governance. Rather than uniform job replacement, the main pattern of role is redefinition under continuous up-skilling.

9.1 Summary of Affected Groups

Entry-level and Beginners job access tightens as automation of code take away may standardized easy job position. Readiness start to improves when candidates start to show AI awareness. Required Skills, AI's Impact, and Up-skilling trend fall upward, while access stabilizes later as AI-first pathways standardize.

Mid-level and graduates are demanded toward a full ownership with AI in the prompting, workflow designing, retrieving and controlling data, testing and monitoring at a scale, and cost and reliability trade-offs. Access improves for people who can demonstrate reliable AI-assisted deliveries.

PhD students and Researcher roles are expanding around the evaluations of science, data governance, and real-world impact. Once standard routine work gets fully powered by AI-assistants, the differentiation, what makes one stand out, will come from originality, credible identification, and monitored, policy aligned deployment.

Lastly, long-term professionals are emphasized to moves toward product ownership, governance, and risk management. Leadership is defined by producing safe outcomes and organizational adoption.

9.2 Final Insights and Message

First, readiness is being modified: high scores for required skills, AI's Impact, and up-skilling across all the different groups show that employability is impated on on orchestration, evaluation, and governance as much as coding. Second, labor polarization is evolving: showcased in figure 14 the temporary gaps followed by the merging at high demand when AI-first practices and hiring signals standardize. Third, human oversight remains irreplaceable: trust, safety, accountability, and standard evaluation long-run value. In summary, AI is not eliminating computer scientists; it is just elevating and redesigning the expectations. Careers will be built on the ability to specify, supervise, and assure AI-enabled systems skills remain central to technical decision-makings.

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